

06_DevelopmentWorkflow.md

MonitorAI v1.5 Development Workflow

Version: 1.5

Status: Baseline

1. Purpose

This document defines the implementation workflow for MonitorAI.

The purpose is to ensure:

- Consistent architecture
 - High code quality
 - Reproducible builds
 - Reliable testing
 - Complete documentation
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2. Development Philosophy

The objective is not:

Writing code quickly.

The objective is:

Building a reliable orthodontic image acquisition platform.

Code quality is more important than development speed.

3. Required Reading Order

Before implementation begins:

Read:

01_MasterSpec.md



02_Workflow_v1.5.md



03_UI_UX_Spec.md



04_FileMakerAPI.md



05_Architecture.md



Remaining Documents

Only then begin implementation.

4. Development Lifecycle

Specification



Implementation



Build Validation



Testing



Documentation Update

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Approval

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Next Phase

5. Implementation Rules

Implement one feature at a time.

Do not start multiple unfinished systems.

Every feature must:

- Compile
- Pass tests
- Be documented

before continuing.

6. Phase 01

Project Foundation

Implement:

- App Entry
- Dependency Injection
- Navigation
- Configuration

Deliverables:

- App Launches
 - Build Clean
-

7. Phase 02

Camera Foundation

Implement:

- AVCaptureSession
- Live Preview
- Torch Control
- Focus Control

Deliverables:

- Stable Camera Preview
 - Clean Build
-

8. Phase 03

Position Calibration

Implement:

- Grid Overlay
- Center Marker
- Retractor Opening Detection
- Stability Detection

Deliverables:

- Calibration Workflow Functional
-

9. Phase 04

Overlay Guide System

Implement:

- PNG Asset Loader
- Overlay Renderer
- Opacity Control

Deliverables:

- Overlay Display Functional
-

10. Phase 05

PTG Foundation

Implement:

- Visibility Scoring
- PTG Dot Logic
- PTG State Management

Deliverables:

- PTG Functional
-

11. Phase 06

Voice Guidance

Implement:

- Voice Engine
- Event Mapping
- Playback Management

Deliverables:

- Voice Guidance Functional
-

12. Phase 07

Haptic Feedback

Implement:

- Taptic Engine Integration
- Event Mapping

Deliverables:

- Haptic Feedback Functional
-

13. Phase 08

Auto Capture

Implement:

- Capture Readiness Detection
- Stability Timer
- Capture Trigger

Deliverables:

- Auto Capture Functional
-

14. Phase 09

Plateau Detection

Implement:

- Plateau Logic
- Plateau Capture

Deliverables:

- Plateau Detection Functional
-

15. Phase 10

Review Workflow

Implement:

- Thumbnail Gallery
- Individual Retake
- Capture Replacement

Deliverables:

- Review Screen Functional
-

16. Phase 11

Metadata Engine

Implement:

- Session Metadata
- Capture Metadata
- Validation

Deliverables:

- Metadata Functional
-

17. Phase 12

FileMaker Integration

Implement:

- Authentication
- Record Creation
- Container Upload
- Metadata Upload

Deliverables:

- Upload Functional
-

18. Phase 13

Offline Queue

Implement:

- Local Storage
- Retry Logic
- Synchronization

Deliverables:

- Offline Upload Functional

19. Phase 14

Analytics

Implement:

- Session Metrics
- PTG Metrics
- Upload Metrics

Deliverables:

- Analytics Functional
-

20. Phase 15

Production Validation

Implement:

- Final Testing
- Bug Resolution
- Documentation Validation

Deliverables:

- Release Candidate
-

21. Build Validation

After every implementation step:

Run:

xcodebuild

Verify:

- Build Success

- No Errors
 - No Warnings
-

22. Error Resolution

Never patch symptoms.

Always identify root cause.

If architecture is incorrect:

Refactor.

Do not create temporary workarounds.

23. Refactoring Rules

Refactoring is encouraged when:

- Duplication appears
- Complexity increases
- Testability decreases

Refactoring is part of development.

Not optional maintenance.

24. Testing Workflow

For every phase:

Unit Tests

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Integration Tests

↓

Manual Validation



Documentation Update

25. Unit Testing Targets

Required:

- PTGService
 - VisibilityScoringService
 - AutoCaptureService
 - PlateauDetectionService
 - MetadataService
 - UploadService
-

26. Integration Testing Targets

Required:

- Camera Workflow
 - Capture Workflow
 - Upload Workflow
 - Offline Queue
 - FileMaker API
-

27. Device Validation

Mandatory:

iPhone 11

iPhone 13

iPhone 15 Pro Max

Latest Supported iPhone

28. Documentation Workflow

When implementation changes:

Update:

- Specification
- Architecture
- API Documentation
- Test Documentation

Code and documentation must remain synchronized.

29. Phase Completion Criteria

A phase is complete only when:

Implementation Complete

Build Successful

Unit Tests Pass

Integration Tests Pass

Documentation Updated

No Critical Issues

30. Definition of Release Candidate

A Release Candidate must satisfy:

Position Calibration Functional

Overlay Functional

PTG Functional

Voice Functional

Haptics Functional

Auto Capture Functional

Plateau Detection Functional

Review Functional

Metadata Functional

FileMaker Upload Functional

Offline Queue Functional

Analytics Functional

31. Final Principle

Every development decision should improve:

- Reliability
- Traceability
- Dataset Quality

Development speed should never be prioritized over dataset quality.